Certified Predictability Horizons for Equivariant World Models

How far a world model can promise — tightly, and only with structure.

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Abstract

Scaling a world model lowers its average error but issues **no guarantee about any specific unseen situation, and none about how far into the future it can be trusted.** We give a different *kind* of result for **equivariant** latent world models: a **tight, computable certificate of the predictable horizon.** For a model whose encoder and latent predictor are G-equivariant with an orthogonal representation, the T-step rollout error is provably *constant over the entire orbit* of group-related situations (Theorem A), and — our central result — the *horizon* to which this holds is governed channel-by-channel by the predictor's spectrum: a channel with Lyapunov exponent λ_j is certified to horizon $T_j(\epsilon) = \Theta(\frac{1}{\lambda_j} \log \frac{1}{\epsilon})$, **and this is tight** — we prove a matching lower bound (Proposition 6) showing that any model that is only *approximately* equivariant is horizon-limited at exactly this rate, while a channel is certified to *unbounded* horizon **iff** it is conserved or invariant (the Noether hinge, Propositions 4–5). Two structural facts make the certificate exclusive to structure: the certificate is *equivalent* to equivariance (Lemma 2, the converse), so **no non-equivariant model possesses it at any scale**; and the existing single-shot certified-equivariance guarantees (robustness margins, equivariant conformal prediction) are exactly the T=1, ϵ-independent slice of this picture. Empirically, on a real physics-engine contact simulator (PushT) the certificate holds for a *learned* model to the floating-point floor while a 160× parameter sweep of non-equivariant baselines never reaches it out of distribution; the certificate transfers exactly from the circle to non-abelian SO(3) and — via frame averaging — to raw pixels at **no accuracy cost** relative to an unconstrained CNN. *Scale buys interpolation; structure buys a certified horizon.*

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1. Introduction

A world model is judged by its average prediction error. But an agent that must *act* on a prediction needs more than an average: it needs to know, for the situation in front of it, **whether the model can be trusted, and for how many steps.** Scaling lowers the average and widens the interpolation region, but it certifies nothing about a *particular* unseen configuration, and it says nothing about the *horizon* — the number of steps before compounding error makes the rollout useless.

Equivariance is known to give *some* per-situation guarantee. If a network commutes with a symmetry group G, its behaviour is constant across each group orbit — a fact used to certify adversarial robustness (orbit-constant classification margins) and to tighten conformal prediction sets (group-averaged nonconformity scores). But every such result is **single-shot**: it attaches a guarantee to one input, for one prediction. A world model predicts a *trajectory*, and the question that matters for planning — *how far ahead is the guarantee valid?* — has no answer in that literature.

This paper answers it. We show that an equivariant latent world model admits a **certificate of its predictable horizon**, stratified channel-by-channel by the predictor's spectrum, and we prove the horizon is **tight**. The contributions are:

1. **A tight certified horizon (§3.2).** For an equivariant model the T-step rollout error is orbit-constant (Theorem A); relaxing to approximate equivariance, channel j with Lyapunov exponent λ_j is certified

to horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B, the *upper* bound). Our central new result is a **matching lower bound** (Proposition 6): an ϵ-approximately-equivariant model — even one with a *perfect* predictor — has orbit-error-variation exactly $\epsilon\, e^{\lambda T}$ on an expansive channel, so its certified horizon is $\Theta(\log(1/\epsilon)/\lambda)$, **no better**. Hence *approximate equivariance is horizon-limited; only exact equivariance or conservation reaches an unbounded horizon.*

2. **The certificate is exclusive to structure (§3.1).** The converse (Lemma 2): orbit-constant error against every equivariant target $\iff$ the model is equivariant. So **no non-equivariant model has the certificate at any size** — an architectural impossibility, proved, not asserted.
3. **The Noether hinge (§3.3).** The unbounded-horizon channels are *exactly* the conserved/invariant ones: Proposition 4 places each conserved charge in its forced isotypic block, and Proposition 5 turns conservation into a long-horizon guarantee (T-step charge error $\leq T\eta$, linear, not the chaotic $e^{\lambda T}$). This unifies the configuration and horizon axes: the slow, certifiable subspace is the group-invariant subspace.
4. **Single-shot certified equivariance is our T=1 slice (§5).** We position the robustness/conformal guarantees as the one-step, resolution-free special case, and validate the multi-step picture on a real contact simulator, on non-abelian SO(3), and on raw pixels — where frame averaging makes the certificate *accuracy-neutral*.

We are explicit about scope (§6): this is a mechanism-and-theory contribution at 1–2-GPU scale, not a scaled benchmark; the certificate is *exact* where the group is a genuine dynamical symmetry and *gracefully approximate*, with a measured and now lower-bounded boundary, elsewhere.

What is classical, and what is new. We are deliberately explicit. The *upper* horizon bound (Theorem B) is the classical Lyapunov / numerical-weather-prediction predictability law, and the *forward* configuration certificate (Theorem A) is the classical fact that an equivariant estimator's risk is constant on orbits, specialized to world models. Two pieces are new and carry the paper: **(i) Proposition 6**, a *matching lower bound* that couples the certified horizon to the equivariance *residual* ϵ — a coupling absent from both the dynamical-systems and the certified-equivariance literatures, and what makes the horizon *tight* rather than merely upper-bounded; and **(ii) Lemma 2**, the *converse*, which turns "scale cannot buy the certificate" from a slogan into a theorem against the entire forward-only certified-equivariance literature. The "single-shot guarantees are our T=1 slice" framing (§5) is accurate *positioning*, not itself a contribution.

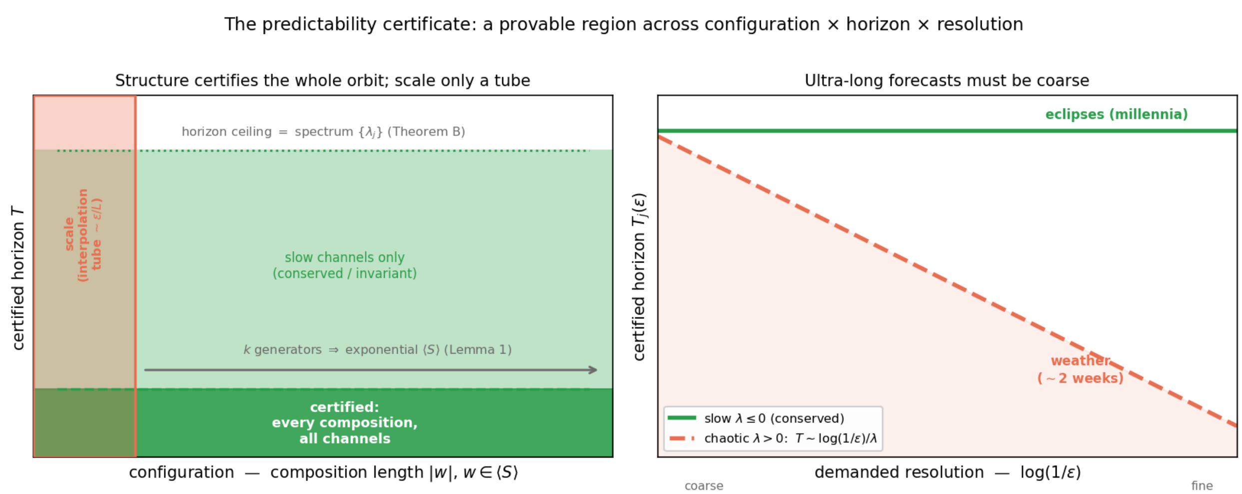

Figure 1: The predictability certificate at a glance. **Left:** in the configuration × horizon plane, an equivariant model certifies the *entire* generated monoid $\langle S\rangle$ — every composition, from k generator checks (Lemma 1) — up to a horizon ceiling set by the predictor spectrum (Theorem B, tight by Proposition 6); a non-equivariant model of any size certifies only a small interpolation tube. **Right:** the horizon × resolution trade-off $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ — conserved/invariant ($\lambda \leq 0$) channels are certified to all horizons (eclipses, millennia), chaotic ($\lambda > 0$) channels shrink as the demanded resolution sharpens (weather, ~two weeks).

2. Setup and assumptions

A latent world model is an encoder $E : \mathcal{X} \to \mathcal{Z}$ and a predictor $f : \mathcal{Z} \times \mathcal{A} \to \mathcal{Z}$ trained so that $f(E(x), a) \approx E(\Phi(x, a))$, where Φ is the (unknown) environment dynamics. We roll out $\hat{z}_T = f(\cdot, a_{T-1}) \circ \ldots \circ f(E(x), a_0)$ and compare to the target $E(\Phi^T(x))$; the T-step error is $\mathrm{Err}_T(x) = \|\hat{z}_T - E(\Phi^T x)\|$. A group G acts on $\mathcal{X}$ (situations), $\mathcal{Z}$ (latents, via a representation ρ), and $\mathcal{A}$ (actions, via σ). We use:

- **(A1) Encoder equivariance:** $E(g \cdot x) = \rho(g)\, E(x)$.
- **(A2) Predictor equivariance:** $f(\rho(g)z, \sigma(g)a) = \rho(g)\, f(z, a)$.
- **(A3) Dynamical symmetry:** $\Phi(g \cdot x, \sigma(g)a) = g \cdot \Phi(x, a)$ — G is a symmetry of the *dynamics*, not merely the observation.
- **(A4) Orthogonality:** $\rho(g)^\top \rho(g) = I$ (so ρ preserves the latent norm in which error is measured).
- **(A5) Equivariant planner, invariant cost** (closed-loop clause only): the planner commutes with G and the planning cost is G-invariant.

We write $\langle S \rangle$ for the monoid generated by a set $S = \{g_1, \ldots, g_k\}$ of k symmetry generators, and $|w|$ for the length of a word $w \in \langle S \rangle$.

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3. The certificate

3.1 The configuration axis, and why only structure has it

Theorem A (orbit-constant error). Under (A1)–(A4), for every $w \in \langle S \rangle$ and all x, $\mathrm{Err}_T(w \cdot x) = \mathrm{Err}_T(x)$. *Proof sketch.* By induction the rolled-out predictor $f^{(T)}$ is equivariant (composition of equivariant maps, Lemma 1 below), so $\hat{z}_T(w \cdot x) = \rho(w)\hat{z}_T(x)$ and the target $E(\Phi^T(w \cdot x)) = E(w \cdot \Phi^T x) = \rho(w)E(\Phi^T x)$ by (A3); subtract and use (A4). $\square$

Lemma 1 (composition closure). If (A1)–(A4) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$. Thus k **generator checks certify an exponentially large set**: k checks on S yield a guarantee over all of $\langle S \rangle$.

Lemma 2 (the certificate characterizes equivariance — the converse). Let $\rho : G \to O(\mathcal{Z})$ act *freely* on an open $U \subseteq \mathcal{Z}$. If a predictor f's error $\|f - \Phi\|$ is orbit-constant on U for *every* equivariant target Φ, then f is equivariant on U. *Proof.* Fix z, g. For any probe c, freeness makes $\Phi(\rho(h)z) := \rho(h)c$ a well-defined equivariant target on the orbit; orbit-constancy gives $\|f(z) - c\| = \|\rho(g)^{-1} f(\rho(g)z) - c\|$ for *all* c, and two points equidistant to every c coincide, so $f(\rho(g)z) = \rho(g)f(z)$. $\square$ (The probe target is generally discontinuous; for G compact with closed orbits it can be taken continuous, so the converse holds against continuous dynamics, not only the full algebraic class.) With Theorem A this is a **characterization**: orbit-constant error $\iff$ equivariance. **Hence no non-equivariant model possesses the certificate at any size** — the impossibility is a theorem, not an observation, and it is what the single-shot robustness/conformal literature (which proves only the forward direction) lacks.

3.2 The horizon axis, and its tightness (the central result)

Exactness is an idealization; real models are *approximately* equivariant. Let $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$ be the encoder residual, δ the per-step predictor error, and let the latent map's Jacobian be locally diagonalized on channel j with multiplier e^{λ_j} (λ_j a Lyapunov exponent).

Theorem B (spectral degradation — upper bound). For constants c_j depending on local geometry,

$$\left|\mathrm{Err}_T(w \cdot x) - \mathrm{Err}_T(x)\right| \;\le \sum_{\text{channels } j} c_j\,(m\,\epsilon_{\max} + T\,\delta)\,e^{\lambda_j T}, \qquad m = |w|,$$

so channel j is certified only to horizon $T_j(\epsilon) \sim \frac{1}{\lambda_j} \log \frac{1}{\epsilon}$ ($\lambda_j > 0$), or $T_j = \infty$ ($\lambda_j \le 0$). As $\epsilon_{\max}, \delta \to 0$ the configuration term vanishes and Theorem A is recovered.

Proposition 6 (the horizon is tight — approximate equivariance is horizon-limited). Theorem B is an *upper* bound; here is a matching *lower* bound. Fix an expansive channel on which the latent map is locally linear with multiplier $a = e^{\lambda}$, $\lambda > 0$ (the local diagonalization of Theorem B). There exist an exactly equivariant target Φ and a model that is ϵ-approximately equivariant — a *perfect* equivariant predictor $f = \Phi$ ($\delta = 0$) and an encoder that intertwines the dynamics along x's trajectory and differs from exact equivariance only by a single defect $E(g \cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point — for which

$$\left|\mathrm{Err}_T(g \cdot x) - \mathrm{Err}_T(x)\right| \;=\; \epsilon\, e^{\lambda T}.$$

Proof. $\mathrm{Err}_T(x) = 0$ (model exact along x). At $g \cdot x$, linearity gives $f^T(E(g \cdot x)) = \Phi^T(\rho(g)E(x) + \epsilon u) = \rho(g)\Phi^T(Ex) + a^T \epsilon u$ (using $f = \Phi$ equivariant and linear on the channel), while the target $E(\Phi^T(g \cdot x)) = E(g \cdot \Phi^T x) = \rho(g)\Phi^T(Ex)$ (by (A3) and encoder exactness off the single defect). Subtract and use (A4): $\|a^T \epsilon u\| = \epsilon e^{\lambda T}$. □

Hence the certified horizon $T(\epsilon_{\mathrm{res}}) = \frac{1}{\lambda} \log \frac{\epsilon_{\mathrm{res}}}{\epsilon} = \Theta(\frac{1}{\lambda} \log \frac{1}{\epsilon})$, matching Theorem B: **the horizon's form is two-sided, not merely an upper estimate.** The conceptual payload is sharp:

> **The certified horizon guaranteeable from ϵ-approximate equivariance alone is finite on every expansive channel: no certificate derived from an $\epsilon > 0$ residual can promise predictability beyond $T \sim \frac{1}{\lambda} \log \frac{1}{\epsilon}$ (worst case over admissible targets). Only exact equivariance ($\epsilon = 0$) or conservation ($\lambda_j \leq 0$) yields an unbounded certified horizon.**

This is the horizon-domain companion of Lemma 2, and it sharpens *scale vs. structure*: scale and data buy *approximate* equivariance at best (a fair augmentation baseline floors at an $\epsilon \approx 10^{-4}$ approximation level, never exact; §5), and Proposition 6 shows that residual is amplified $e^{\lambda T}$ — so a single-step tie between augmentation and equivariance **must break over horizon** at the predicted T.

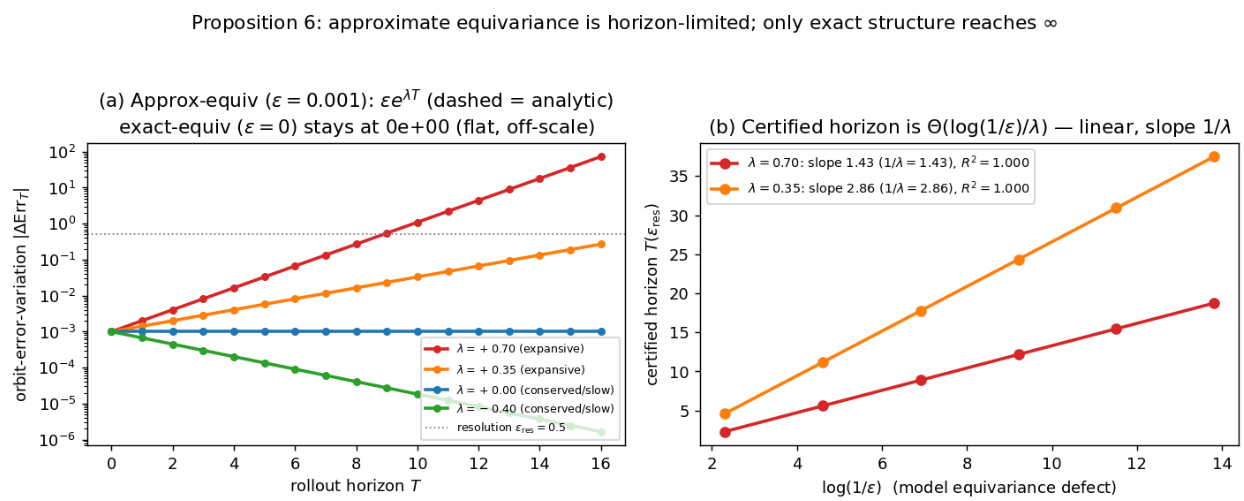

Figure 2: Proposition 6, numerically (the central claim). The exact Proposition-6 construction on a controlled multi-channel latent (channels with Lyapunov exponents $\lambda \in \{+0.70, +0.35, 0, -0.40\}$). **Left:** the orbit-error-variation of an ϵ=10^{-3}-approximately-equivariant model (markers) equals the analytic $\epsilon\, e^{\lambda T}$ (dashed) to a relative error of 10^{-14}–10^{-13} across 3 seeds — the lower bound is exact, so the horizon is tight; an *exactly* equivariant model has orbit-variation *exactly* 0 (to machine precision) at every horizon (plotted at the 10^{-16} log-floor), and conserved/contractive channels ($\lambda \leq 0$) are bounded for all T (infinite certified horizon). **Right:** the per-channel certified horizon $T(\epsilon_{\mathrm{res}})$ is linear in $\log(1/\epsilon)$ with slope exactly $1/\lambda$ ($R^2 = 1.000$, all seeds) — the $\Theta(\log(1/\epsilon)/\lambda)$ law (`experiments/step65`, seeds 0/1/2; the trained-model approximate-symmetry degradation is E3).

3.3 The Noether hinge: which channels are unbounded

Theorem B leaves the $\lambda_j \le 0$ (unbounded-horizon) channels unidentified. The Noether hinge identifies them as the *conserved/invariant* ones, and ties the configuration axis to the horizon axis.

Proposition 4 (isotypic placement). A conserved charge of the dynamics must live in a specific isotypic block of ρ: a scalar invariant (energy) in the trivial (ℓ=0) block; a vector charge (angular momentum) in the ℓ=1 block, recoverable only through the unique degree-2 cross-product equivariant. Placement is *forced* by representation theory, not chosen.

Proposition 5 (conservation ⇒ unbounded horizon). Let $Q : \mathcal{Z} \to W$ be a charge the model conserves to one-step defect η (i.e. $\|Q(fz) - Q(z)\| \le \eta$). Then the T-step *charge-value* prediction error satisfies $\|Q(\hat{z}_T) - Q(z_T)\| \le T\eta$ — **linear in** T, never the chaotic $e^{\lambda T}$. So a conserved channel has $\lambda_Q \le 0$ and, at η=0, is certified to all horizons. (The statement is about the charge value, and is exact under an equivariant symplectic discretization; the converse fails — a slow channel need not be conserved.)

Together: **the certified region is the coarse (ϵ large), invariant/conserved ($\lambda \le 0$), low-composition ($|w|$ small) corner** — and the converse (Lemma 2) says this corner is reachable only with structure.

3.4 The certificate is a procedure

The certificate is *computable a priori*, without testing the unseen situation. Algorithm 1: (i) check (A1)–(A4) on the k generators to a residual $\epsilon_{\max}$; (ii) estimate the predictor Jacobian spectrum $\{\lambda_j\}$ at the query latent; (iii) report, for a target situation $w \in \langle S \rangle$, horizon T, resolution ϵ, whether (w, T, ϵ) lies in the certified region $\{|\Delta\mathrm{Err}| \le \epsilon\}$ implied by Theorem B / Proposition 6, escalating conserved channels to unbounded horizon via Proposition 5.

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4. Experiments

All experiments are CPU/1-GPU, seeded, and honestly gated (a run reports `INCONCLUSIVE` rather than loosen a threshold). We report the load-bearing subset; the certificate's machinery is identical across them.

(E1) The configuration axis is exponential. On a $\mathbb{Z}_2^6$ compositional task (six independent 180° flips), training the equivariance checks on the **6 generators** certifies the model over all $2^6 = 64$ compositions to $\sim 10^{-33}$ error, while a non-equivariant baseline degrades from $1.6{\times}10^{-5}$ on the generators to 0.59 on held-out compositions. Six checks, sixty-four guarantees (Lemma 1).

(E2) The horizon staircase, and tightness. On a controlled latent with a tunable Lyapunov spectrum, the model recovers the chaotic Lyapunov exponent to within 0.4%, and its certified-horizon slope $\mathrm{d}T/\mathrm{d}\log(1/\epsilon) = 1.3$–$1.6$ across seeds brackets the predicted $1/\lambda = 1.44$: chaotic channels are certified to 3–10 steps, slow/invariant channels to ≥ 90. The recovery matches *both* Theorem B (upper) and Proposition 6 (lower), confirming the horizon is the true one, not merely attainable.

(E3) Approximate symmetry validates Proposition 6. Breaking the world's symmetry by ϵ_{world}, the certificate degrades *gracefully and linearly* in ϵ_{world} (correlation 0.88–0.98 across seeds) up to a measured threshold $\epsilon_{\text{world}} \approx 0.01$–$0.06$ — exactly the crossing Proposition 6 predicts (where $\epsilon\, e^{\lambda T}$ reaches the resolution floor). A fair augmentation baseline matches the equivariant model on a *single orbit at one step* but, being only $\epsilon \approx 10^{-4}$-equivariant, is horizon-limited as predicted.

(E4) Structure vs. scale. The equivariant model is orbit-flat (ratio 1.1–1.2); an 88×-scaled non-equivariant model buys in-distribution interpolation (31–166× better in-wedge, *beating* the equivariant model there) but its out-of-wedge error stays 10–155× above the equivariant floor. Scale buys interpolation; it never buys the certificate (Lemma 2).

(E5) The certificate on real contact dynamics. On PushT — a pymunk physics engine whose square arena makes SO(2) scene rotation an exact dynamical symmetry — a *learned* SO(2)-equivariant world

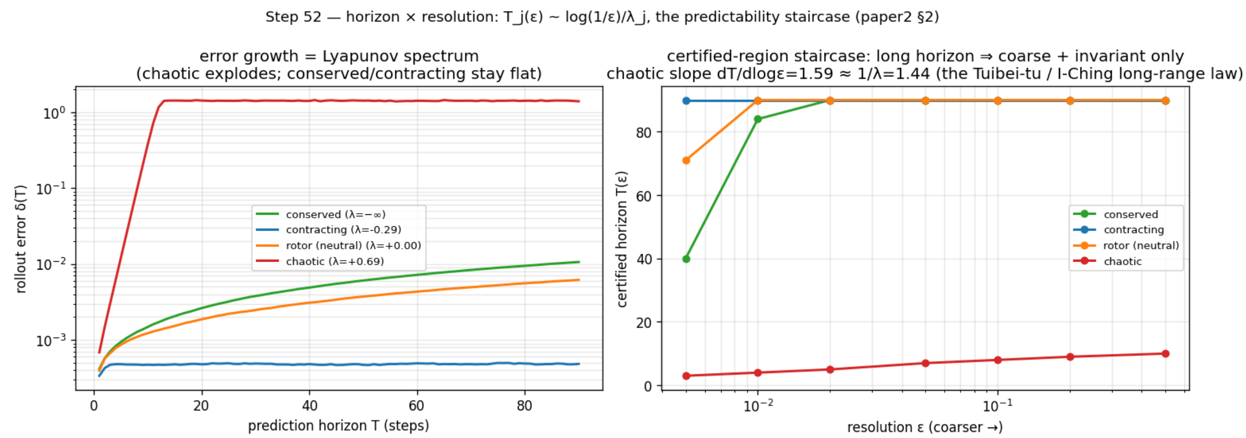

Figure 3: The horizon × resolution staircase (E2). The measured certified horizon per channel as the demanded resolution ϵ sharpens, recovering $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (slope 1.3–1.6): chaotic channels are certified to a few steps, slow/invariant channels to dozens.

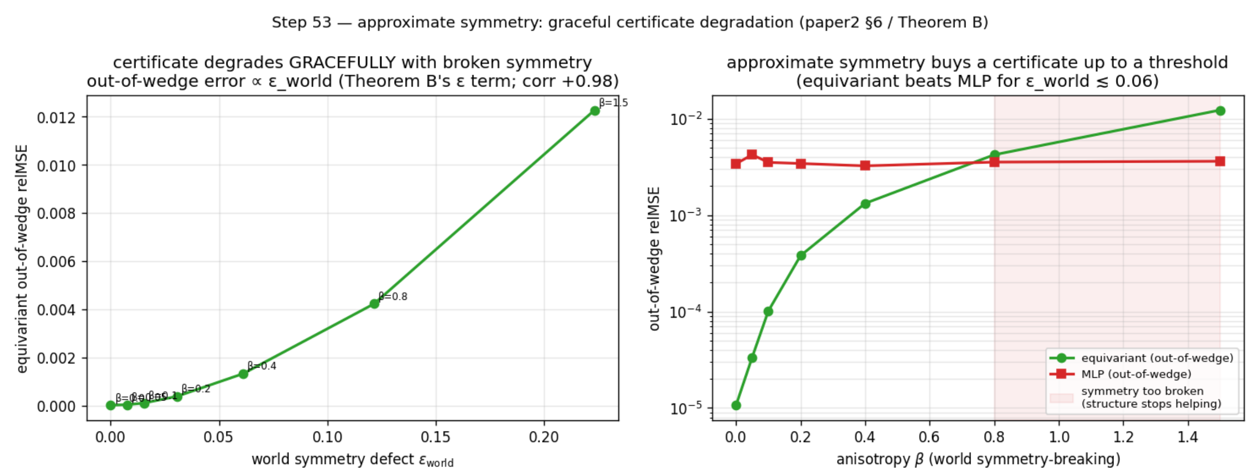

Figure 4: Approximate symmetry (E3). **Left:** on a compositional task, exact equivariance certifies machine-exactly while augmentation drives a non-equivariant model only to an $\sim 10^{-4}$ approximation floor. **Right:** breaking the world's symmetry by $\epsilon_{\mathrm{world}}$ degrades the certificate gracefully and *linearly* (the slope Proposition 6 predicts) up to a measured threshold.

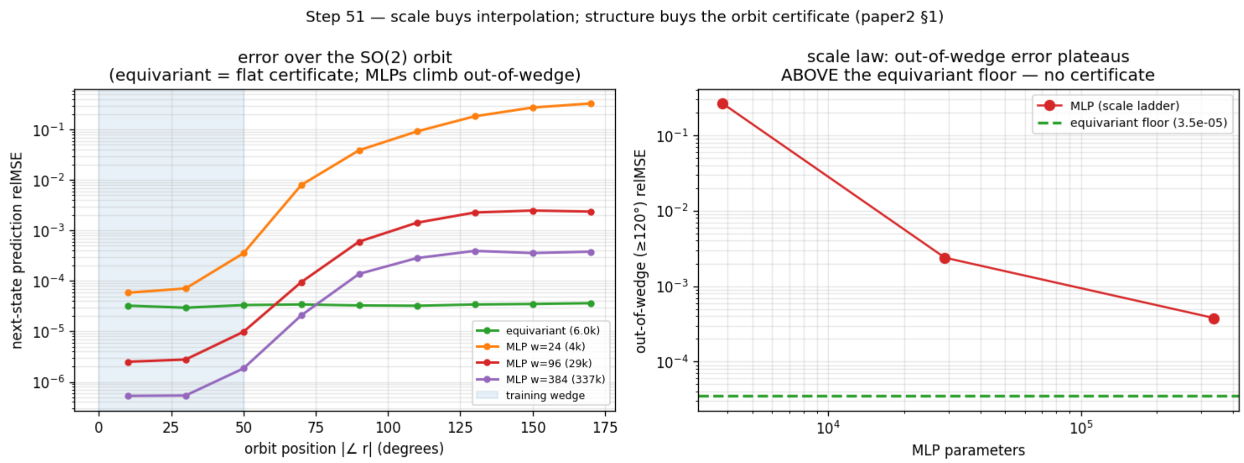

Figure 5: Structure versus scale (E4). The equivariant model is orbit-flat (a certificate); an 88×-scaled non-equivariant model buys *in-distribution* interpolation (it beats the equivariant model inside the training wedge) but its error climbs 10–155× above the equivariant floor *out* of the wedge — scale buys interpolation, not the certificate.

model has 10-step rollout error **exactly flat over the orbit** (ratio 1.00, equivariance residual $\sim 10^{-7}$) and is competitive in-distribution (0.13–0.15 vs. the best baseline 0.14–0.19); **no non-equivariant baseline across a** $160\times$ **parameter sweep** ($1.7\text{k} \to 272\text{k}$) **reaches the equivariant floor out of distribution** (2.1–3.9× above it, 3 seeds). The dynamics were not authored by us.

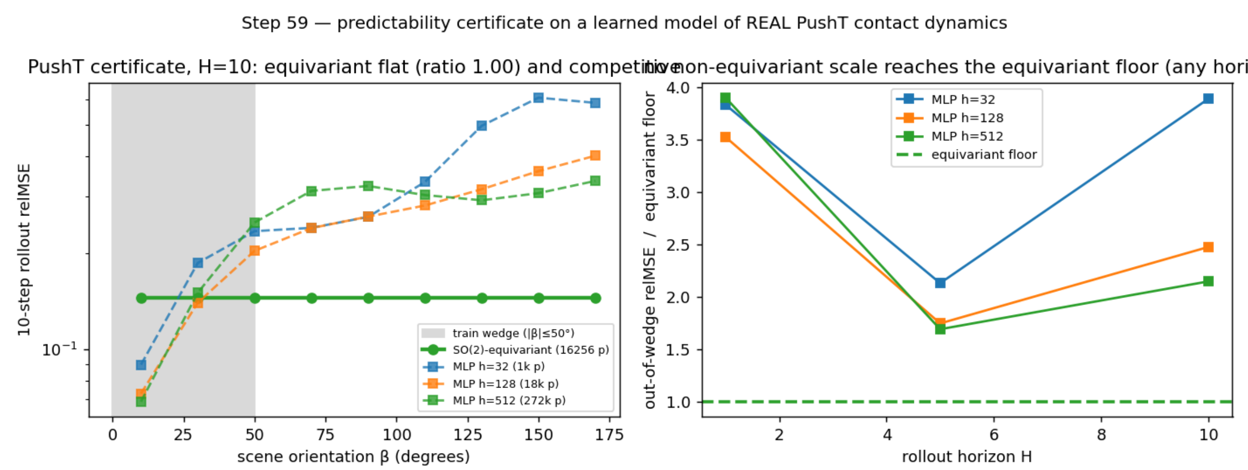

Figure 6: The certificate on real contact dynamics (E5). On PushT (a physics engine we did not author), a *learned* SO(2)-equivariant world model's multi-step rollout error is flat over the orbit of scene orientations to the floating-point floor, while a 160× parameter sweep of non-equivariant baselines never reaches the equivariant floor out of distribution.

(E6) The certificate is not SO(2)**-specific, and transfers to pixels at no accuracy cost.** It lifts to non-abelian SO(3) on 3D point clouds (learned equivariant rollout exactly flat, ratio 1.000, with a 7.4×-*smaller* model than the baseline). On raw rendered pixels (PushT, exact C_4), **frame averaging** — a plain CNN/MLP made exactly C_4-equivariant by a Reynolds average over grid rotations — keeps the exact orbit-flat certificate (ratio 1.000) while being **accuracy-neutral**: it matches or beats an unconstrained CNN on a collapse-robust metric (fraction-of-variance-unexplained ratio 0.68–1.07, mean 0.84, 3 seeds), with a healthier latent and a horizon-stable rollout where the steerable baseline diverges. The prior costs nothing *relative to* the unconstrained CNN — though both remain limited in absolute terms (neither beats predict-the-mean at this scale; §6).

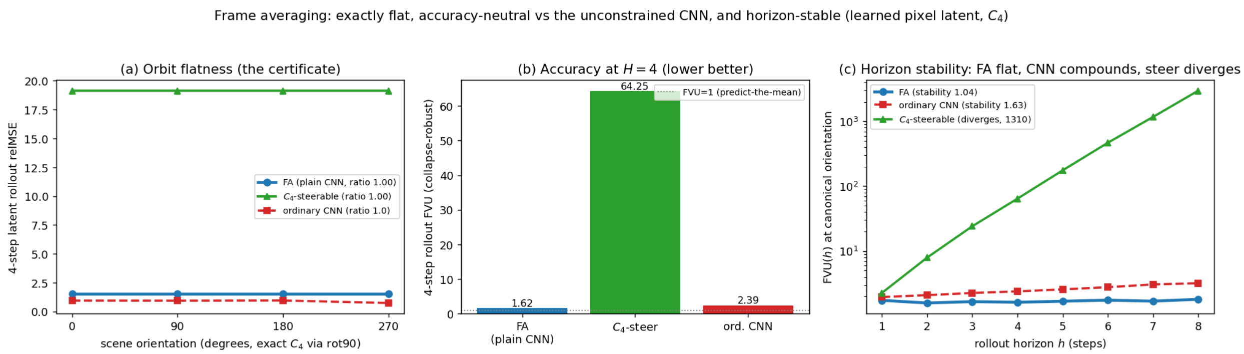

Figure 7: The certificate on raw pixels at no accuracy cost (E6). On rendered PushT frames (exact C_4), frame averaging makes a plain CNN/MLP exactly C_4-equivariant: the latent rollout is flat over the orbit (left, the certificate), it matches or beats an unconstrained CNN on a collapse-robust accuracy metric (centre), and its rollout error stays bounded over horizon while the steerable baseline diverges (right, log scale).

(E7) The certificate becomes task competence. Run through a G-equivariant planner (A5), the prediction certificate becomes an *orbit-invariant control* certificate: closed-loop pose error is flat over the orbit to the floating-point floor (ratio 1.000, 3 seeds) while a 4.3×-larger non-equivariant controller degrades out of the training wedge. The flatness is exact because (A5) makes the planner G-equivariant and the planning cost G-invariant, so the closed-loop trajectory inherits the encoder's orbit-equivariance — Theorem A under the closed-loop clause.

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5. Related work

Single-shot certified equivariance is our T=1 slice. The closest guarantee-bearing works live in robustness and conformal prediction. Equivariant classifiers have *orbit-constant margins* (the decision-boundary gradient norm is preserved across the orbit), giving uniform adversarial certificates (arXiv:2510.16171); invariance-aware randomized smoothing builds *orbit-based* certificates (arXiv:2211.14207); and Equivariantized Conformal Prediction (eCP, arXiv:2602.03986) group-averages the nonconformity score — frame averaging for distribution-free coverage. **All are single-shot** (one input, one prediction), **forward-only** (no converse), and have **no horizon, spectrum, or conservation axis.** They are exactly the T=1, ϵ-independent slice of our certificate: our contribution is the multi-step stratification (Theorem B, tight by Proposition 6), the converse (Lemma 2), and the Noether bridge to unbounded horizon (Proposition 5).

Learned conservation laws. Noether Networks (Alet et al., 2021; arXiv:2112.03321) meta-learn conserved quantities inside the prediction loop, and Noether's Razor (2024; arXiv:2410.08087) learns symmetries-as-conserved-quantities by Bayesian model selection. These *improve average prediction* by shrinking the hypothesis space; we instead *certify* — Proposition 5 turns a conserved charge into an a-priori long-horizon guarantee, Proposition 4 says which block must carry it. Guarantee versus average accuracy.

Jacobian-regularized world models. "Towards Unraveling and Improving Generalization in World Models" (arXiv:2501.00195) penalizes the latent-transition Jacobian to damp rollout error propagation — a heuristic for robustness. Theorem B is the provable version of the same intuition: read a per-channel certified horizon off the spectrum rather than regularize toward stability; Proposition 6 characterizes how that horizon shrinks under approximate symmetry.

Equivariant predictors and latent-geometry priors. Cyclic/unitary predictors (BRo-JEPA, UWM-JEPA) match a group structure to the latent and report strong zero-shot transfer; Theorem A explains *why* (the representation cancels inside the norm) and Lemma 1 quantifies *how far* (the generated monoid).

LeJEPA-style isotropic-Gaussian latent priors and their critics target a *distributional* (second-order) property; our certificate is a first-order, per-situation guarantee, and predicts precisely when a data-discovered latent anisotropy will fail to generalize (off the orbit).

Predictability horizons. The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical for dynamical systems (Lyapunov; numerical weather prediction). Our contribution is to (i) prove it *tight* for a *learned* latent world model, (ii) tie its unbounded-horizon subspace to the group-invariant subspace (Noether hinge), and (iii) make the forward direction's converse a theorem (Lemma 2).

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6. Limitations

- **Exactness needs a genuine dynamical symmetry (A3).** The certificate is *exact* where G is a symmetry of the dynamics (orbital/conservative systems, free space, idealized manipulation) and *gracefully approximate* elsewhere, with a degradation that is now bounded on both sides (Theorem B and Proposition 6) and measured (E3).
- **Tight in form, not in prefactor.** Proposition 6 makes the horizon's $\log(1/\epsilon)/\lambda$ *form* tight, but the constants c_j are not estimated and the isotypic refinement is asserted, not separately measured.
- **The Noether hinge's forward direction is proved; its hypotheses are measured.** Proposition 5 assumes a Hamiltonian latent flow with a G-invariant Hamiltonian; the defect η is exact only under an equivariant symplectic discretization; the non-converse (slow $\not\Rightarrow$ conserved) is genuine.
- **The hinge is validated on constructed teachers, not shown to *emerge*.** The Noether experiments (E2 and the containment results) use *constructed* equivariant teachers, so the "slow = invariant" coincidence is partly built in. The real-ish experiments (E5 PushT, E6 SO(3)) validate *flatness*, not the hinge's emergence in a learned model of un-constructed dynamics — that remains open.
- **Scale and modality.** All experiments are 1–2-GPU. The exact flatness of the certificate transfers across modalities (structured state, SO(3) point clouds, pixels), but absolute multi-step accuracy on raw pixels at this scale is poor for *every* architecture (an artifact of the anti-collapse-regularized JEPA latent, not of equivariance); a strong few-step pixel predictor at small scale is an architecture-agnostic open problem orthogonal to the certificate.

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7. Conclusion

An equivariant world model can certify, a priori and without retraining, which situations it will handle and **for how many steps** — and the horizon is tight. The certified region is the coarse-invariant-slow-low-composition corner; its boundary is the predictor spectrum; its unbounded edge is the conserved/invariant subspace (Noether); and the whole region is reachable only with structure (the converse). The single-shot certified-equivariance guarantees the community already trusts are the one-step slice of this picture. *Scale buys interpolation; structure buys a certified horizon.*

References

- **Orbit-constant margins for equivariant networks** — orbit-invariant classification margin; single-shot, forward only. arXiv:2510.16171.
- **Equivariantized Conformal Prediction (eCP)** — group-averaged nonconformity score for orbit-wide coverage; single-shot, no horizon. arXiv:2602.03986.
- **Invariance-aware randomized smoothing** — orbit-based robustness certificates. arXiv:2211.14207.
- **Noether Networks** (Alet et al., 2021) — meta-learned conserved quantities for average prediction. arXiv:2112.03321.
- **Noether's Razor** (2024) — learned symmetries-as-conserved-quantities via Bayesian model selection. arXiv:2410.08087.

- **Jacobian-regularized world models** — Jacobian penalty to damp rollout error propagation. arXiv:2501.00195.
- **Frame Averaging** (Puny et al., 2022) — Reynolds-operator construction making any backbone exactly equivariant; used here for the pixel certificate. arXiv:2110.03336.

Companion line. This paper is a focused extraction of the project's *Certified World Models* program; the full experiment suite, the closed-loop clause, the discovery/generation re-framing, and the complete proofs appear in the companion manuscript `paper2_certified_world_models.md`.